

Problem Statement ID - 26181

Problem Statement Title - A secure AI health companion for real-time monitoring, early risk detection, and disaster resilience through on-device intelligence.

Theme - MedTech / BioTech / HealthTech

PS Category - Hardware

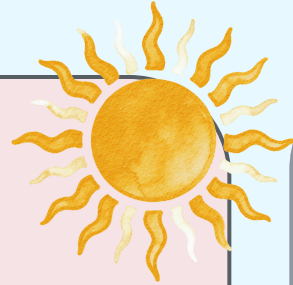
Team Name - Braj Coders

Team ID - 1111

Team Members - Ritika Rawat, Sahil Chaudhary, Sweety Yadav, Nikhil Singh, Pawan Kumar, Shubham Jain



554 Days



In 2024, India experienced a **significant rise in heatwave days**, reaching 554 compared to 230 in 2023.

75%



of India's workforce, accounting for nearly 380 million people, is involved in outdoor and other heat-exposed occupations, making them particularly vulnerable to rising temperatures and extreme heat conditions.

50/100

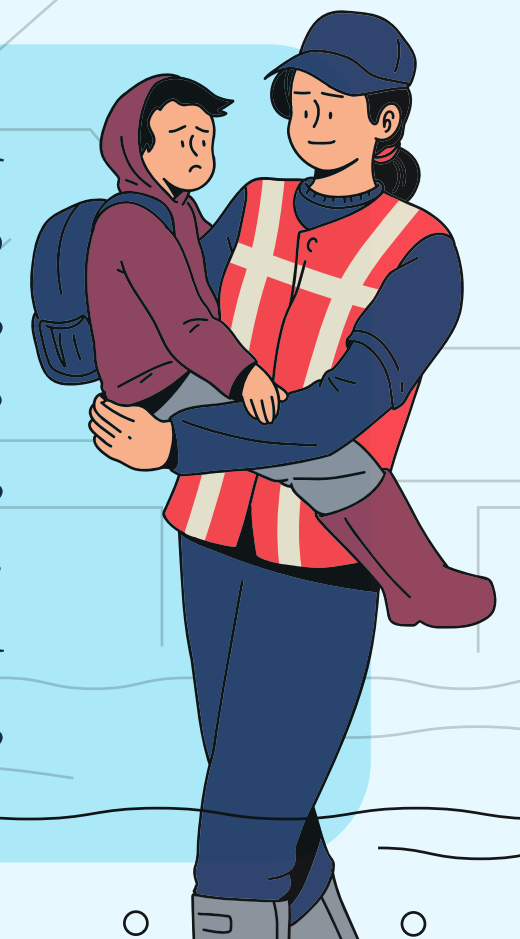
rural internet penetration vs 127/100 in urban India the exact gap where cloud-dependent health apps fail rural India sits at **just 48.31 per cent...** Urban India, by contrast, stands at **126.80 per 100 population.**



2024

was the warmest year on record in the country since 1901, and climate-related hazards keep intensifying. Most wearable health tech assumes **constant internet access** the exact thing that breaks down **during the disasters** this problem is about.

India faces recurring health emergencies during heat waves, floods, and pollution events - heat stress, dehydration, and cardiac complications spike, especially among elderly citizens, outdoor workers, and rural populations. **Healthcare access is often delayed** exactly when disasters disrupt roads, power, and connectivity.



GENERIC FITNESS BANDS (Mi Band / Fire-Boltt class)



Track basic heart rate via PPG
(optical pulse, not real ECG)

Gap: No real ECG, no on-device AI, no offline SOS, no environmental or disaster-specific awareness.

PREMIUM WEARABLES (Apple Watch class)



Offer limited ECG with capable hardware.

Gap: Requires a paired phone + active data connection. No air-quality or disaster-specific alerts.

CLOUD-BASED HEALTH APPS



Offer AI-driven health insights and risk scoring.

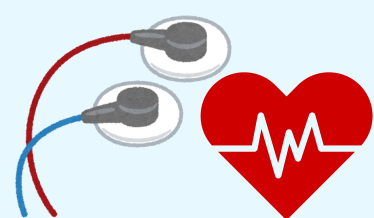
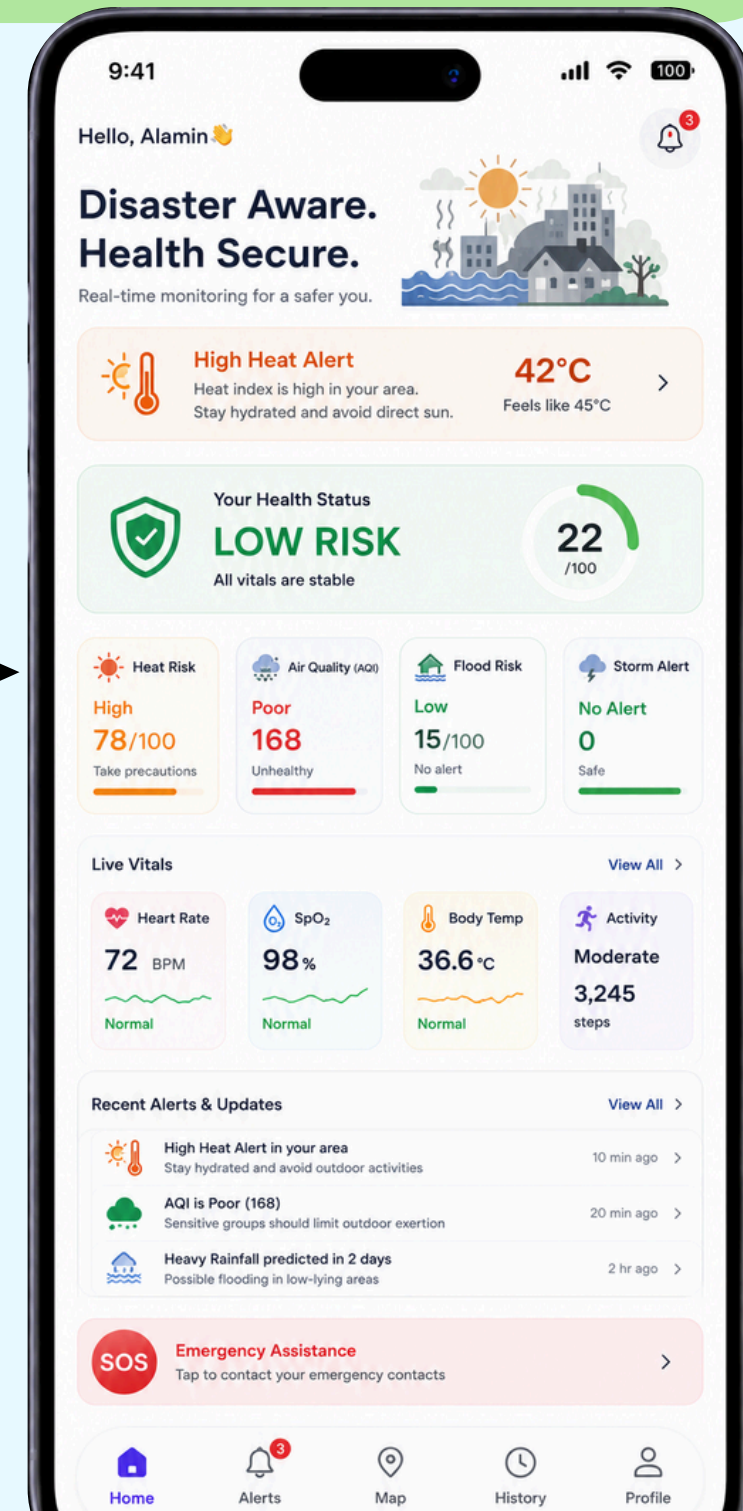
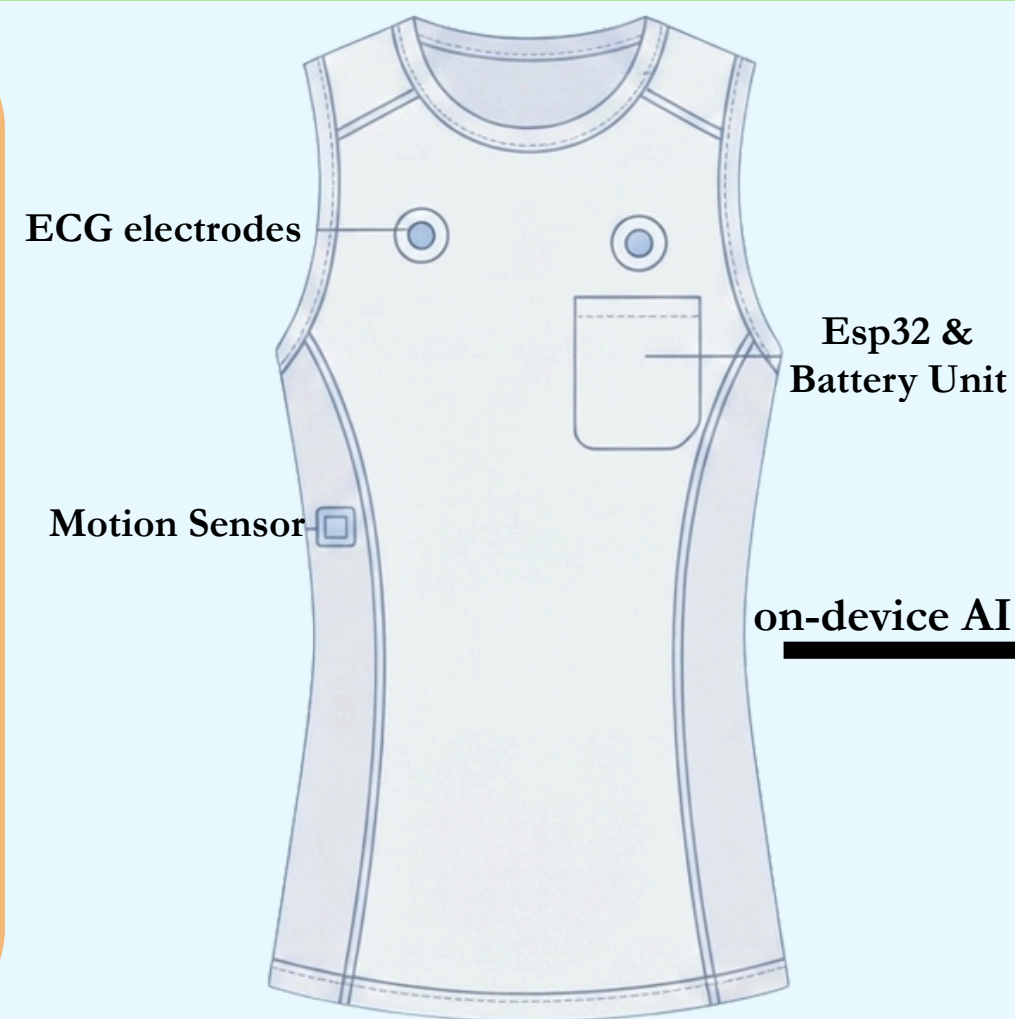
Gap: Depend on continuous internet and send raw health data to servers — creating privacy and reliability risks.

Evaluating key features

Features	Generic Fitness Band (Mi Band/Fire-Boltt class)	Apple Watch/Premium Wearable	Cloud-Based Health Apps
Real ECG (not just PPG)	✗	✓ (limited, requires manual activation)	✗
On-device AI (no cloud)	✗	Partial	✗ AI runs on cloud
Works fully offline (SOS)	✗	⚠ (needs paired phone + data)	✗ Requires internet
Air quality sensing	✗	✗	✗
Disaster-specific alerts (heat/AQI)	✗	✗	✗
Cost	Low-Medium	Very High	Low-Medium

“Develop a wearable that uses on-device AI to spot health risks early and stay useful during disasters.” SIH26181
Sanjeevni reads your heartbeat, your movement, and the air around you, and warns you before things turn serious. All on your phone. Nothing sent to the cloud. Built for India's heat waves, floods, and pollution.

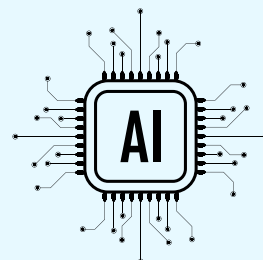
- ✓ Real heart signal, not guesswork: an actual ECG sensor, not just a pulse count.
- ✓ Every alert is decided on your phone, not a server somewhere.
- ✓ Knows the difference between "you're just exercising" and "something's wrong".
- ✓ Still works with no internet: sends an SOS by SMS if needed.
- ✓ Built into your clothing, tracks your rest and sleep too, so you never have to remember to wear a separate device.



Real ECG hardware



Runs on your phone,
not the cloud



Reads context,
not just numbers



Works 100%
Offline



Daily Wearable,
not a gadget



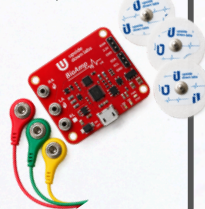
Personalizes over time

TIER 1 - SENSING LAYER

BioAmp (ECG/PPG/EMG)



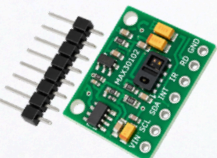
- Heart Rhythm
- HRV, Arrhythmia detection



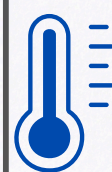
MAX30102,



- Heart rate
- Blood oxygen saturation



DHT22 / BME280



- Temperature + Humidity
- Heat index
- Dehydration risk



MQ135 / PM2.5 Sensor



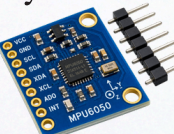
- Air Quality
- Pollution
- AQI index



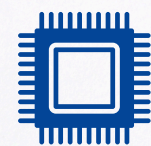
MPU6050



- Accelerometer + Gyro
- Fall detection
- Activity tracking



Environment Node



Sensor Integration Unit

ESP32



Sensor Hub

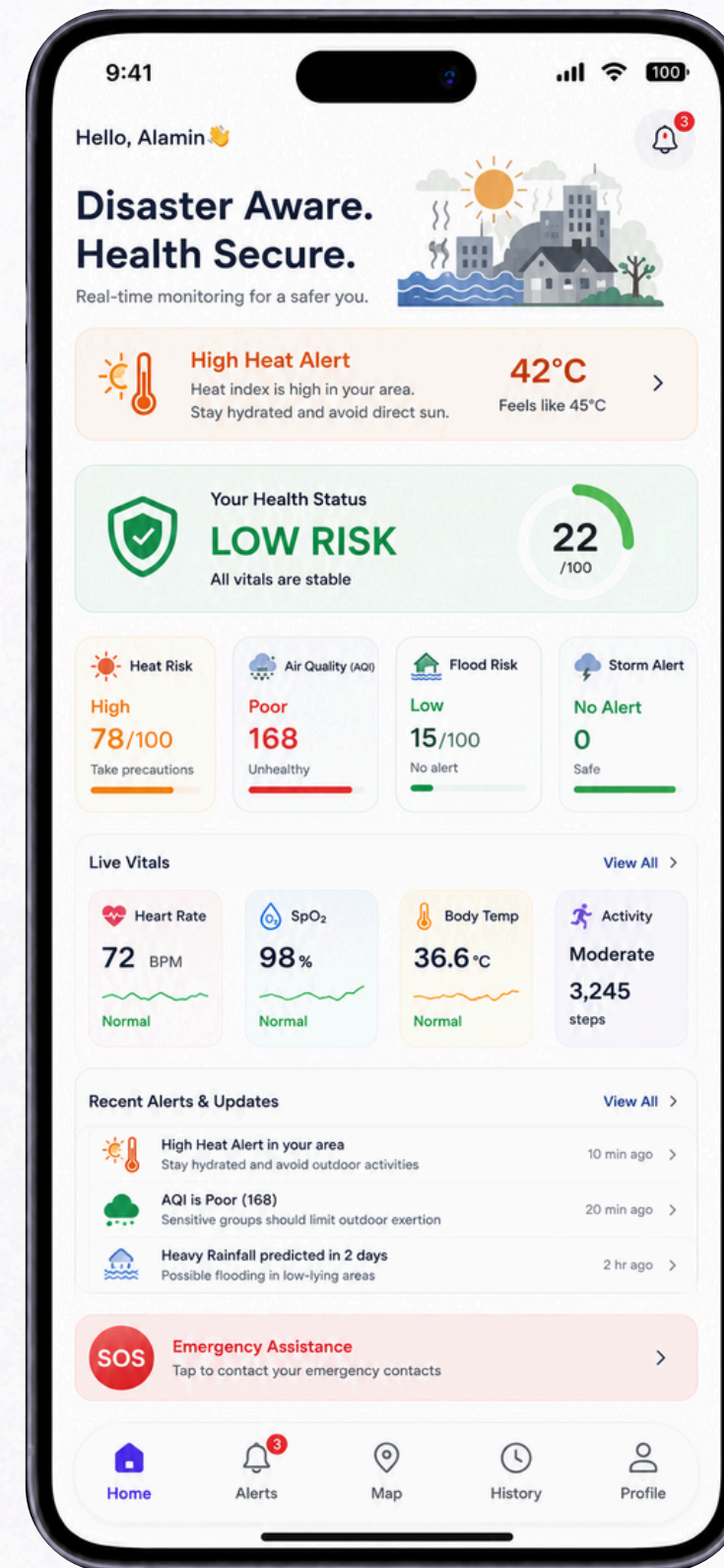
- Data acquisition
- Basic filtering
- Feature extraction
- BLE transmission



Motor / Actuator Control Unit

- Haptic Feedback
- Alert Vibration
- Buzzer / LED

TIER 2- EDGE AI INFERENCE LAYER



HR:
72 BPM

SpO₂:
98%

Temp:
32 °C

AQI:
45

STATUS:
NORMAL

TIER 3- USER LAYER/ THE WEARABLE



ON DEVICE AI MODELS (TensorFlow Lite)

Fall Detection Model

Heart Health Model

Respiratory Risk Model

Heat Stress Model

ALERTS & ACTIONS



Location



Local Alert

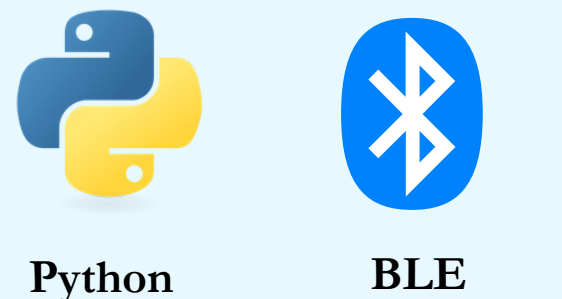
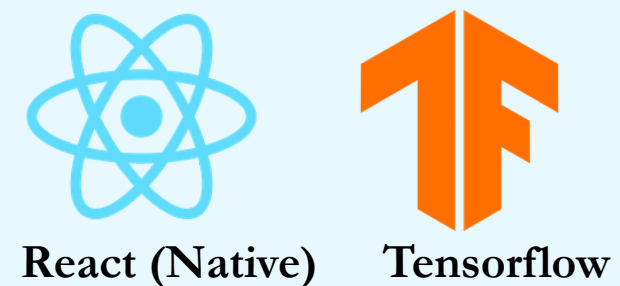


SOS Call

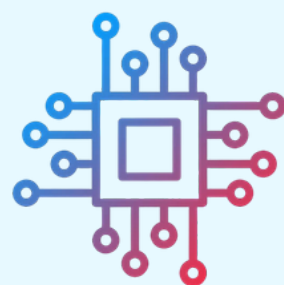


Works without internet

TECHSTACK



SCALABILITY



Modular Hardware Design

Sensor modules (ECG, AQI, motion) can be added/removed independently, letting the product scale from a **3-sensor MVP** to a **full 5-sensor version** without redesigning the core system



Low-Cost Manufacturing

Sub-₹2,500 unit cost enables mass deployment across population-scale use cases, **not just premium/niche markets**



Platform-Agnostic Software

On-device TFLite models run on any modern Android phone via NNAPI — no dependency on specific hardware, so the **app scales across device tiers**



Geographic Scalability

Fully offline operation means the product scales into rural and low-connectivity regions **without needing new infrastructure**

SUSTAINABILITY



Zero Recurring Cloud Cost

On-device inference means **no per-user server/API cost** — the business model stays financially sustainable even at large scale



Low Power Design

Duty-cycled sampling and low-power BLE extend battery life, **reducing charging frequency and long-term energy consumption**



Open-Source Stack

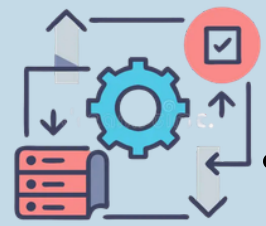
Built on **React Native, TensorFlow Lite, ESP32** — avoids licensing lock-in, keeping long-term maintenance costs low



Reusable, Durable Hardware

Passive undergarment sensing (not disposable) **supports long-term reuse** rather than a consumable/replaceable product model

CURRENT LIMITATIONS: HARDWARE



- **Dry-electrode ECG** signal can degrade with heavy movement or sweat mitigated but not eliminated
- **Prototype** currently validates 3 of 5 sensors in the full product vision; body temperature and sleep are inferred, not directly measured yet

CURRENT LIMITATIONS: BATTERY & USAGE



- **Continuous multi-sensor sampling** affects battery life over all-day wear.
- Garment need **consistent compression fit for reliable electrode** contact comfort/reliability tradeoff still being tuned

CURRENT LIMITATIONS: SCOPE



- **No clinical/diagnostic validation** yet positioned as a **risk-alerting tool**, not a medical device
- **Not yet tested across diverse body types, skin tones**, or extended real-world wear conditions

Where We Are Now

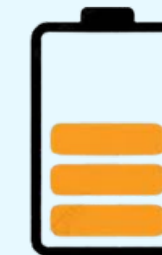
Garment currently validated on 3 sensors (ECG, motion, air quality)



Single garment form factor, individual use only



On-device rule-based and statistical anomaly detection



English-only interface, single-user deployment



Where We're Headed

Expand to full 5-sensor coverage add SpO2 and dedicated ambient thermal sensing

Extend to smartwatch/band form factors and add caregiver/family dashboard view

Deploy trained CNN-based ECG arrhythmia classifier (MIT-BIH based) for deeper cardiac pattern detection

Regional-language voice alerts, ASHA-worker/PHC distribution channel for rural-scale rollout